Is Grokking a Loss of Normal Hyperbolicity of the Interpolation Manifold?

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Abstract: A recent line of work recasts the post-memorization phase of grokking as constrained optimization: once a network interpolates the training set, weight decay drives a slow drift along the zero-loss manifold toward lower norm. In the language of dynamical systems, this is a fast–slow system in which the interpolation manifold plays the role of a slow manifold. We ask a question that this framing makes natural but the existing literature does not address: *is the sharp generalization transition a loss of normal hyperbolicity of that manifold*: a fold- or bifurcation-like event in which a normal restoring direction goes flat? Or does the manifold stay uniformly attracting while generalization happens by smooth drift? We propose a simple, optimizer-agnostic diagnostic: the smallest nonzero singular value $\sigma^{+}_{\min}(\mathbf{J})$ of the residual Jacobian, which, for the squared loss, equals the slowest normal restoring rate of the manifold. On a two-layer ReLU network trained to grok modular addition under squared loss, $\sigma^{+}_{\min}(\mathbf{J})$ does not collapse at the transition; it is near zero only *before* memorization and attains its largest values *during* the transition. The result holds across five seeds, and the six smallest singular values behave identically; there is no subspace-local collapse either. This is preliminary evidence against the bifurcation hypothesis and in favor of the smooth-contraction picture. We are explicit that a single-setting, gradual-transition experiment under `Adam` optimizer does not prove the absence of a bifurcation; it constrains where one could hide.

Keywords: grokking, normal hyperbolicity, interpolation manifold, fast–slow dynamics, implicit bias, weight decay, Gauss–Newton Jacobian, delayed generalization

1 Introduction

Grokking is the empirical observation that a network can reach perfect training accuracy with near-chance test accuracy and only generalize after a long additional period of training [Po22]. It is often discussed alongside *double descent*, in which test error first falls, then rises as the model overfits, and eventually falls again [Be19; Na21]. The two are related but not identical: Power et al. [Po22] distinguished grokking on the grounds that its generalization occurs far past the interpolation threshold, while Davies et al. [DLK22] argue both arise from a single pattern-learning-speed mechanism. Our question is orthogonal to that taxonomy: we ask about the *geometry* of the grokking transition rather than its classification.

A productive recent account treats the long post-memorization phase as constrained optimization. Muşat [Mu25] argues that, after interpolation, gradient descent with small weight decay effectively minimizes the parameter norm subject to staying on the zero-loss

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manifold, in the limit of infinitesimal learning rate and weight decay; a concurrent framework reaches the same endpoint through a Riemannian norm flow on the manifold of interpolating solutions [BPD25]. Lyu et al. Lyu et al. [Ly24] prove, for homogeneous networks with large initialization and small weight decay, a sharp transition from a kernel predictor to a KKT point of a global min-norm/max-margin problem [LL20; So18] on both classification and regression, with modular addition under cross-entropy as the motivating phenomenon.

All of these share a structure: a fast process pulls the parameters onto a low-dimensional set of (near-)interpolating solutions, and a slow process driven by weight decay moves the parameters *along* that set. This is exactly a singularly perturbed, or fast-slow, dynamical system, with the interpolation manifold as the slow manifold. Yet none of these works invokes the corresponding mathematics: normal hyperbolicity, Fenichel persistence of slow manifolds [Fe79; Ku15], and what happens when normal hyperbolicity is *lost*. We ask whether grokking's defining feature, the suddenness of the transition, is a signature of that loss.

Concretely, we contrast two hypotheses. **(H1)** The interpolation manifold loses normal hyperbolicity at the transition: a normal restoring direction softens toward zero, the fast/slow timescale separation breaks, and the trajectory jumps: a fold or pitchfork in the fast–slow decomposition. Modular addition's symmetry group makes a symmetry-breaking pitchfork a natural candidate normal form. **(H0)** The manifold stays uniformly normally hyperbolic; the transition is a feature of the slow drift through a region where test behavior changes quickly, without any degeneration of the manifold's geometry. This is the picture implied by the contraction-based accounts above. If **H1** holds true, this would tie the dynamical and mechanistic-interpretability [Na23] views together. Our contribution is a clean test that distinguishes H1 from H0, and a preliminary, and as it turns out, negative result.

2 The diagnostic

Let $f(\boldsymbol{\theta}; \mathbf{x}) \in \mathbb{R}^c$ be the network output and $L(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \|f(\boldsymbol{\theta}; \mathbf{x}_i) - \mathbf{y}_i\|^2$ the squared loss on a training set of size n. We use squared loss precisely because it makes the *interpolation manifold* $\mathcal{M} = \{\boldsymbol{\theta} : f(\boldsymbol{\theta}; \mathbf{x}_i) = \mathbf{y}_i \ \forall i\}$ an exact object with a well-defined tangent and normal structure. Training with decoupled weight decay λ follows, in the gradient-flow limit, $\dot{\boldsymbol{\theta}} = -\nabla L(\boldsymbol{\theta}) - \lambda\boldsymbol{\theta}$; for small λ the term $-\nabla L$ is fast and pulls toward $\mathcal{M}$, while $-\lambda\boldsymbol{\theta}$ is the slow drift along it.

Collect residuals into $\mathbf{r}(\boldsymbol{\theta}) = (f(\boldsymbol{\theta}; \mathbf{x}_i) - \mathbf{y}_i)_i \in \mathbb{R}^{nc}$ and let $\mathbf{J}(\boldsymbol{\theta}) = \partial r/\partial\boldsymbol{\theta} \in \mathbb{R}^{nc\times D}$ be the residual Jacobian. The Gauss–Newton matrix is $\mathbf{G} = \mathbf{J}^\top\mathbf{J}$. At a point of $\mathcal{M}$ the row space of $\mathbf{J}$ is the normal space $N_\theta\mathcal{M}$ (directions in which residuals, hence the loss, change to first order), and $\ker \mathbf{J}$ is the tangent space $T_\theta\mathcal{M}$ (directions along which the model stays interpolating). The nonzero eigenvalues of $\mathbf{G}$ are the squared singular values $\sigma_i(\mathbf{J})^2$, so the smallest normal restoring rate is

$$\kappa(\boldsymbol{\theta}) = \sigma^+_{\min}\big(\mathbf{J}(\boldsymbol{\theta})\big)^2,$$

where $\sigma^+_{\min}$ is the smallest nonzero singular value. For an over-parameterized network with $nc < D$, $\mathbf{J}$ has generically full row rank, so all nc singular values are positive and $\sigma^+_{\min}$ is their minimum.

Why $\sigma^+_{\min}$ captures normal contraction. Near an interpolating point $\boldsymbol{\theta}^\star$ (so $\mathbf{r}(\boldsymbol{\theta}^\star) = 0$) the fast flow is $\dot{\boldsymbol{\theta}} = -\nabla L(\boldsymbol{\theta})$ with $\nabla L = \frac{2}{n}\mathbf{J}^\top\mathbf{r}$. Its linearization is governed by the Hessian $\nabla^2 L = \frac{2}{n}\big(\mathbf{J}^\top\mathbf{J} + \sum_k r_k \nabla^2 r_k\big)$, whose second term *vanishes at* $r = 0$; hence $\nabla^2 L(\boldsymbol{\theta}^\star) = \frac{2}{n}\mathbf{J}^\top\mathbf{J}$, the Gauss–Newton matrix. Writing $\partial\boldsymbol{\theta} = \partial\boldsymbol{\theta}_\parallel + \partial\boldsymbol{\theta}_\perp$ along $T_{\boldsymbol{\theta}^\star}\mathcal{M} = \ker\mathbf{J}$ and $N_{\boldsymbol{\theta}^\star}\mathcal{M} = \operatorname{row}(\mathbf{J})$, the linearized fast dynamics is $\dot{\partial\boldsymbol{\theta}} \approx -\frac{2}{n}\mathbf{J}^\top\mathbf{J}\,\partial\boldsymbol{\theta}$: tangential components lie in $\ker\mathbf{J}$ and are not restored (the slow directions), while normal components contract at the rates $\frac{2}{n}\sigma_i(\mathbf{J})^2$. The slowest normal contraction rate is therefore $\propto \sigma^+_{\min}(\mathbf{J})^2$. Normal hyperbolicity of $\mathcal{M}$ is exactly the statement that this rate is bounded away from zero uniformly along the slow drift – the condition under which Fenichel theory guarantees the perturbed slow manifold persists and the reduced flow is the norm flow of [BPD25; Mu25]. A loss of normal hyperbolicity, the prerequisite for the fold/pitchfork of H1, is exactly $\sigma^+_{\min}(\mathbf{J}) \to 0$. The diagnostic is thus the single scalar trajectory $\sigma^+_{\min}(\mathbf{J}(\boldsymbol{\theta}(t)))$ read against test accuracy: a dip toward zero at the transition supports H1; a value bounded away from zero supports H0. Since $\sigma^+_{\min}(\mathbf{J})$ depends only on the model and data at θ, it is well defined under any optimizer, not only the flow for which the slow-drift theory is cleanest.

3 Experimental setup

We train a two-layer ReLU network $f(\boldsymbol{\theta};\mathbf{x}) = \mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)$ with width 96 and no second-layer bias on modular addition mod $p = 11$: the input concatenates the one-hot encodings of a and b, and the target is the one-hot encoding of $(a + b) \bmod p$. We use a random 70%/30% train/test split, squared loss, full-batch optimization, and a large initialization (Kaiming ×3.5) to induce a clear memorization plateau [LMT23]. To reach the transition within a practical compute budget, we optimize with AdamW (learning rate 3×10^{-3}, decoupled weight decay 2.0) for 35,000 steps. We snapshot $\boldsymbol{\theta}$ every 500 steps and compute $\sigma^+_{\min}(\mathbf{J})$ (via the SVD of $\mathbf{J}$, in `float32`) in a separate pass, so the decomposition does not slow training. Because ReLU makes $\mathbf{J}$ piecewise constant in the activation pattern, $\mathbf{J}$ can change across activation switches; in practice the resulting $\sigma^+_{\min}$ trajectory is smooth at snapshot resolution. We repeat the run over five initialization seeds.

4 Results

The run groks (Fig. 1): training accuracy reaches 1.0 by about step 4,000; test accuracy sits at 0.0, *below* the ≈ 0.09 chance level for 11 classes, i.e. the memorizing solution actively mispredicts test points, through about step 6,000, then rises over roughly steps 7,000–17,000 to ≈ 0.95 before settling near 0.86.

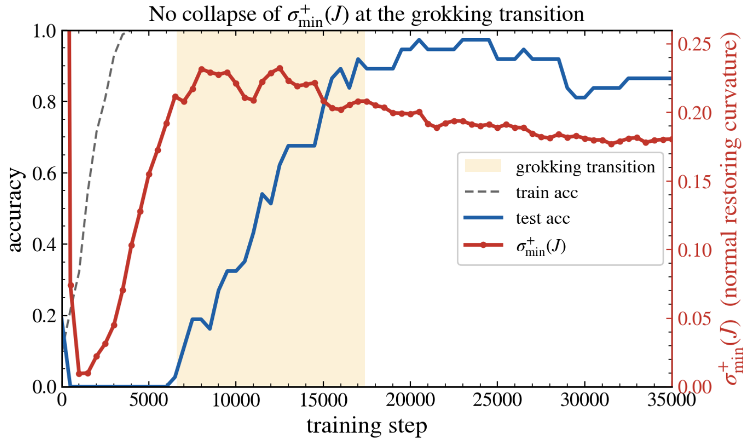

Fig. 1: Test/training accuracy (left axis) and the normal restoring rate $\sigma^+_{\min}(\mathbf{J})$ (right axis). The shaded band marks the generalization transition. $\sigma^+_{\min}(\mathbf{J})$ does not dip during the transition; it is small only before memorization and largest while the model generalizes. Two-layer ReLU network, squared loss, AdamW, modular addition $p = 11$; single seed.

Phase	**Step range**	**median $\sigma^+_{\min}(\mathbf{J})$**	**min $\sigma^+_{\min}(\mathbf{J})$**
Pre-memorization	500–1800	≈ 0.010	0.0096
Memorization plateau	3000–6000	0.128	0.045
Generalization transition	7000–17000	0.220	0.202
Post-transition	≥ 22000	0.182	0.177

Tab. 1: Normal restoring rate by training phase (single seed). The only regime in which $\sigma^+_{\min}(\mathbf{J})$ approaches zero is pre-memorization, which is unrelated to grokking; during the transition it is at its maximum. The pre-memorization reading also shows the diagnostic is sensitive enough to register a genuinely near-degenerate Jacobian when one occurs.

The diagnostic does not behave as **H1** predicts. $\sigma^+_{\min}(\mathbf{J})$ is near zero only *before* memorization: its global minimum, $\approx 1 \times 10^{-2}$, occurs at step 1,000, while training accuracy is still well below 1 and the network has not yet differentiated its outputs. It then rises monotonically as the manifold forms, and across the entire transition it sits at its largest values; the normal restoring rate *peaks* mid-transition rather than collapsing. Table 1 summarizes the medians by phase.

Two checks address the natural objections that a single number or a single seed could mislead (Fig. 2). First, the *six* smallest singular values behave identically: they form a tight cluster that stays bounded away from zero throughout the transition, so the absence of a dip is not an artifact of looking only at the global minimum, no low-dimensional subspace collapses either (panel a). Second, the result is robust across five seeds: although

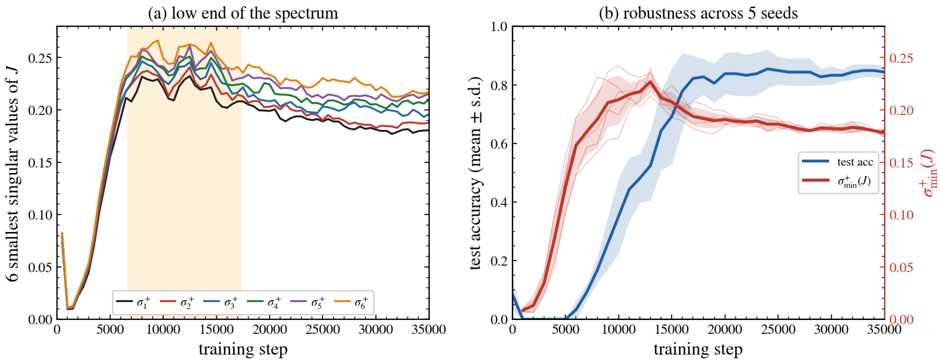

Fig. 2: (a) The six smallest singular values (σ_1^+ being the smallest) of $\mathbf{J}$ form a tight cluster bounded away from zero across the transition; the smallest dips only pre-memorization. The yellow shaded region identifies the grokking transition. (b) Robustness across five seeds: thin red lines are per-seed $\sigma_{\min}^+(\mathbf{J})$, the bold red curve and band are mean ± s.d., and the blue curve/band is mean ± s.d. test accuracy. No seed shows a dip at its transition.

the transition occurs at different steps per seed (test accuracy first exceeds 0.5 between steps $\approx$ 10,000 and 16,000), in every seed $\sigma_{\min}^+(\mathbf{J})$ during its transition window remains in the 0.18–0.23 range, and the seed-averaged curve shows no dip (panel b).

A secondary observation: the parameter norm does not decrease through the transition in this run (it stays near 10 and drifts slightly upward), so generalization is not accompanied by visible ℓ_2-norm contraction here. This is consistent with AdamW's implicit bias being governed by an ℓ_∞-type rather than ℓ_2 geometry [GV26; XL24], and we treat it as a caveat about the optimizer rather than a claim about the manifold. We refrain from reporting the dynamical timescale ratio $\sigma_{\min}^+(\mathbf{J})^2/\lambda$: for the gradient-flow theory λ is the drift rate, but AdamW's decoupled weight decay does not play that role (Adam rescales the effective step per coordinate), so the raw ratio would misstate the separation; a clean ratio belongs to a gradient-flow study.

5 Discussion and limitations

Taken at face value the experiment favors **H0** over **H1**: the interpolation manifold remains robustly normally hyperbolic across grokking, and the suddenness of the transition is not mirrored by any softening of a normal restoring direction. This is the outcome expected from the smooth-contraction picture of [BPD25; Ly24; Mu25] and from recent work framing the grokking delay as exponential norm contraction across a geometric gap: Truong et al. [Tr26] derive a norm-separation delay law[2] for discrete regularized SGD. Because the entry–exit

[2] We cite it for context and have not independently reproduced it.

delay law of a delayed-loss-of-stability mechanism only applies if a bifurcation exists, the negative result also removes the motivation for that route.

We are deliberate about what this does not establish. First, we optimize with AdamW, whereas the fast–slow theory is stated for gradient descent or gradient flow; the sharpest grokking regimes (GD with large initialization and small weight decay) are untested here. Second, the observed transition is moderately gradual rather than maximally sharp. Third, $\sigma^{+}_{\min}(\mathbf{J})$ (and the six smallest) is still a global measure; a loss of normal hyperbolicity confined to a narrow, test-relevant subspace (for instance the symmetry-aligned directions of modular addition) could in principle occur while the low end of the spectrum stays bounded away from zero, and a subspace-resolved diagnostic would be needed to exclude it. Fourth, we measure a discrete trajectory, not the gradient-flow limit in which the manifold is exactly defined. We therefore frame the result as constraining, not refuting: in this regime, grokking is not accompanied by a global collapse of the manifold's normal restoring rate.

These limitations also name the experiments that would settle the question: the same diagnostic under gradient descent in a sharp-transition regime; a subspace-resolved curvature projected onto the directions that most separate train from test points (or onto symmetry-aligned subspaces); and, if a dip never appears, a positive characterization of which geometric quantity *does* move monotonically through the transition, which under AdamW may be an ℓ_∞- or sparsity-based measure rather than ℓ_2 norm.

6 Conclusion

We asked whether the sharp transition in grokking is a loss of normal hyperbolicity of the interpolation manifold, and gave a diagnostic: the smallest nonzero singular value of the residual Jacobian that turns the question into a single measurable curve, justified for the squared loss as the slowest normal restoring rate. On a network that groks modular addition under squared loss, the answer appears to be no, robustly across seeds and across the low end of the spectrum: the manifold stays normally hyperbolic through the transition. We do not claim to have proved the absence of a bifurcation; we show that, in a clean grokking run with a sensitive instrument, no such signature is present, which is itself a constraint on theories of why the transition is sharp.

Reproducibility. Entire training and diagnostic code, data behind Fig. 1 and 2, and all the necessary scripts are available at GitHub[3].

[3] GitHub Repo: https://github.com/baksho/grokking-normal-hyperbolicity

References

[Be19] Belkin, M. et al.: Reconciling Modern Machine-Learning Practice and the Classical Bias–Variance Trade-off. Proceedings of the National Academy of Sciences 116 (32), pp. 15849–15854, 2019.

[BPD25] Boursier, E.; Pesme, S.; Dragomir, R.-A.: A Theoretical Framework for Grokking: Interpolation followed by Riemannian Norm Minimisation. 39th Conference on Neural Information Processing Systems (NeurIPS 2025), 2025, arXiv: 2505.20172, https://arxiv.org/pdf/2505.20172.

[DLK22] Davies, X.; Langosco, L.; Krueger, D.: Unifying Grokking and Double Descent. In: NeurIPS 2022 Machine Learning Safety Workshop. 2022.

[Fe79] Fenichel, N.: Geometric Singular Perturbation Theory for Ordinary Differential Equations. Journal of Differential Equations 31 (1), pp. 53–98, 1979, https://www.sciencedirect.com/science/article/pii/0022039679901529.

[GV26] Gronich, E.; Vardi, G.: The Implicit Bias of Adam and Muon on Smooth Homogeneous Neural Networks. 2026, arXiv: 2602.16340, https://arxiv.org/pdf/2602.16340.

[Ku15] Kuehn, C.: Multiple Time Scale Dynamics. Springer, 2015.

[LL20] Lyu, K.; Li, J.: Gradient Descent Maximizes the Margin of Homogeneous Neural Networks. In: International Conference on Learning Representations (ICLR). 2020, arXiv: 1906.05890, https://openreview.net/pdf?id=SJeLIgBKPS.

[LMT23] Liu, Z.; Michaud, E. J.; Tegmark, M.: Omnigrok: Grokking Beyond Algorithmic Data. In: International Conference on Learning Representations. 2023, arXiv: 2210.01117, https://openreview.net/pdf?id=zDiHoIWa0q1.

[Ly24] Lyu, K. et al.: Dichotomy of Early and Late Phase Implicit Biases Can Provably Induce Grokking. In: The 12th International Conference on Learning Representations (ICLR). 2024, arXiv: 2311.18817, https://proceedings.iclr.cc/paper_files/paper/2024/file/909c8fef63e1cede406ce9e6794f99a2-Paper-Conference.pdf.

[Mu25] Musat, T.: The Geometry of Grokking: Norm Minimization on the Zero-Loss Manifold. arXiv preprint arXiv:2511.01938, 2025, https://arxiv.org/abs/2511.01938.

[Na21] Nakkiran, P. et al.: Deep Double Descent: Where Bigger Models and More Data Hurt. Journal of Statistical Mechanics: Theory and Experiment 2021, 2021.

[Na23] Nanda, N. et al.: Progress Measures for Grokking via Mechanistic Interpretability. In: The 11th International Conference on Learning Representations (ICLR). 2023, arXiv: 2301.05217, https://arxiv.org/pdf/2301.05217.

[Po22] Power, A. et al.: Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. 1st Mathematical Reasoning in General Artificial Intelligence Workshop, 2022, arXiv: 2201.02177, https://arxiv.org/abs/2201.02177.

[So18] Soudry, D. et al.: The Implicit Bias of Gradient Descent on Separable Data. Journal of Machine Learning Research 19 (70), pp. 1–57, 2018, https://www.jmlr.org/papers/volume19/18-188/18-188.pdf.

[Tr26] Truong, K. X. et al.: The Norm-Separation Delay Law of Grokking: A First-Principles Theory of Delayed Generalization. 2026, arXiv: 2603.13331, https://arxiv.org/pdf/2603.13331.

[XL24] Xie, S.; Li, Z.: Implicit Bias of AdamW: ℓ_∞-Norm Constrained Optimization. In: International Conference on Machine Learning (ICML). 2024, arXiv: 2404.04454, https://arxiv.org/pdf/2404.04454.